

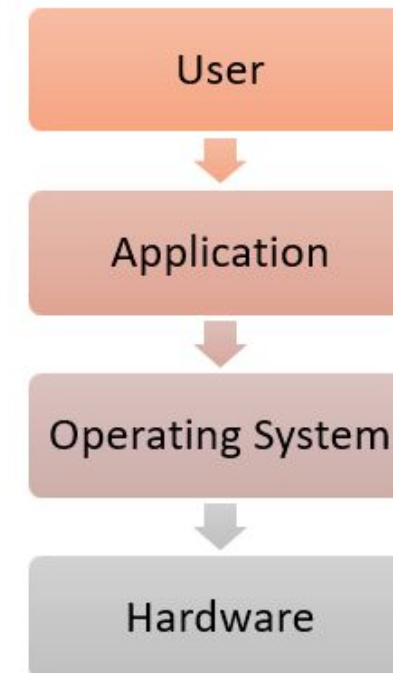
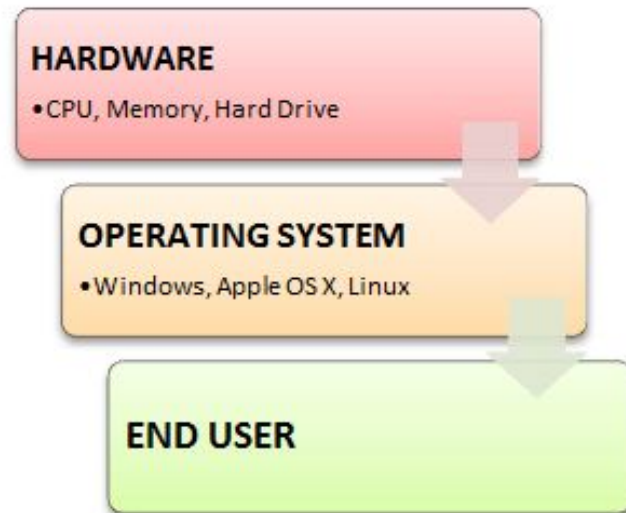
A large, faint watermark of the ISTAD logo is centered in the background. It is a circular emblem with a blue outer ring containing the text 'INSTITUTE OF SCIENCE AND TECHNOLOGY ADVANCED DEVELOPMENT' and Khmer text. Inside the ring is a blue circle with binary code '01010011 01010100 01000001 0100100' around its perimeter. The center features a stylized atom with three orbits and a blue speech bubble with three dots.

DevOps

Operating System Concept

What is Operating System

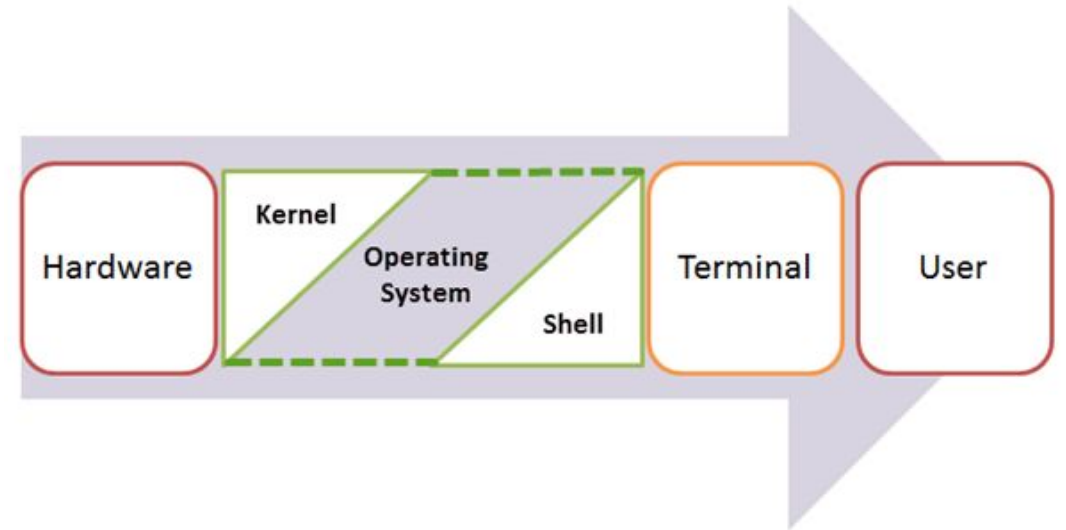
- **An operating system (OS)** is a software program that acts as an *intermediary* between a **user** and a **computer** hardware. It provides a platform to run other application programs and performs tasks such as **memory management**, **process management**, and **security**. Some of the well-known operating systems are Microsoft Windows, Apple macOS, and Linux.



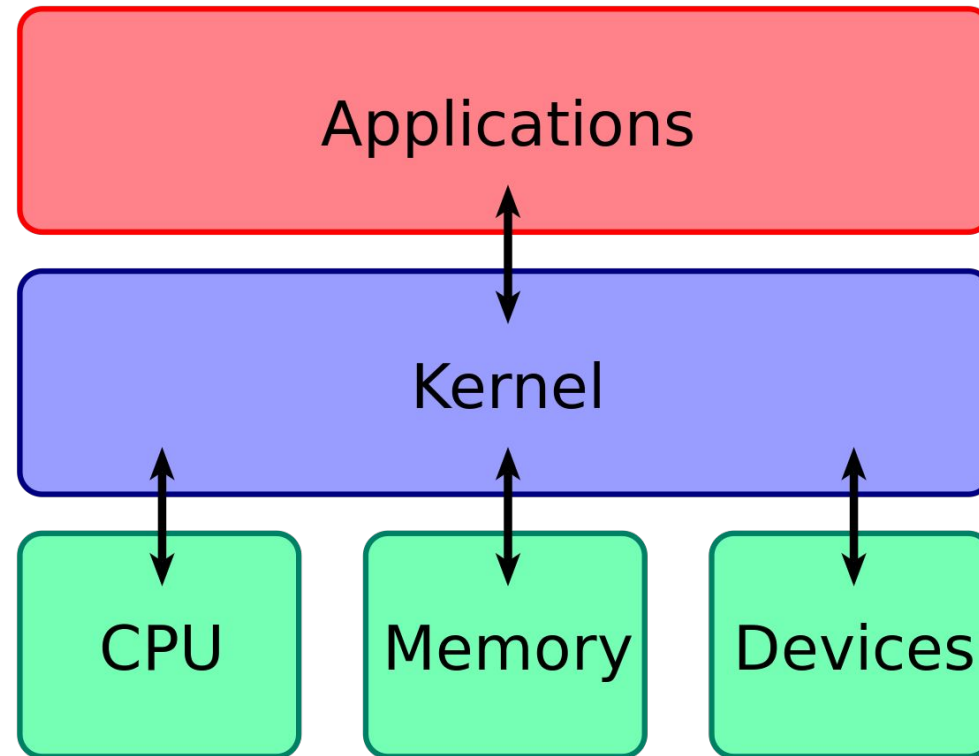
What is Kernel?

- A **kernel** is the central component of an operating system that provides the basic functions for all other parts of the operating system to build upon. The kernel acts as a bridge between the hardware and the software applications running on the system.

It controls the allocation of resources, such as **memory, processing time**, and **input/output operations**, and manages **interactions** between the applications and the underlying hardware.



What is Kernel?



An oversimplification of how a kernel **connects** application software to the hardware of a computer

What is Kernel?

❖ **Linux kernel :**

- **Linux Distributions:** Ubuntu , Fedora, Debian, CentOS, and many others
- **Embedded System:** such as routers, smartphones , industrial automation system to home appliances.
- **Real-time System:** in case that it has to respond to events or input within a short time frame, they must be reliable and working correctly. These system are important for area like avionics (aircraft tech) and telecommunication , where everything happens quickly and without error.

What is Kernel?

❖ **Microsoft Windows NT kernel:**

- Window Version : WindowNT, Windows 2000, Windows XP, Windows Vista....., Window 11
- Server Operating System:
- Window Subsystem for Linux (WSL): provides layer for running Linux binary executable on Windows.

What is Kernel?

❖ **Mach kernel (Apple macOS):**

- macOS and iOS:
- Darwin: open source core of macOS
- NeXSTEP: originally develop for the NeXSTEP operating system, which was precursor to macOS.

Functions of Operating System



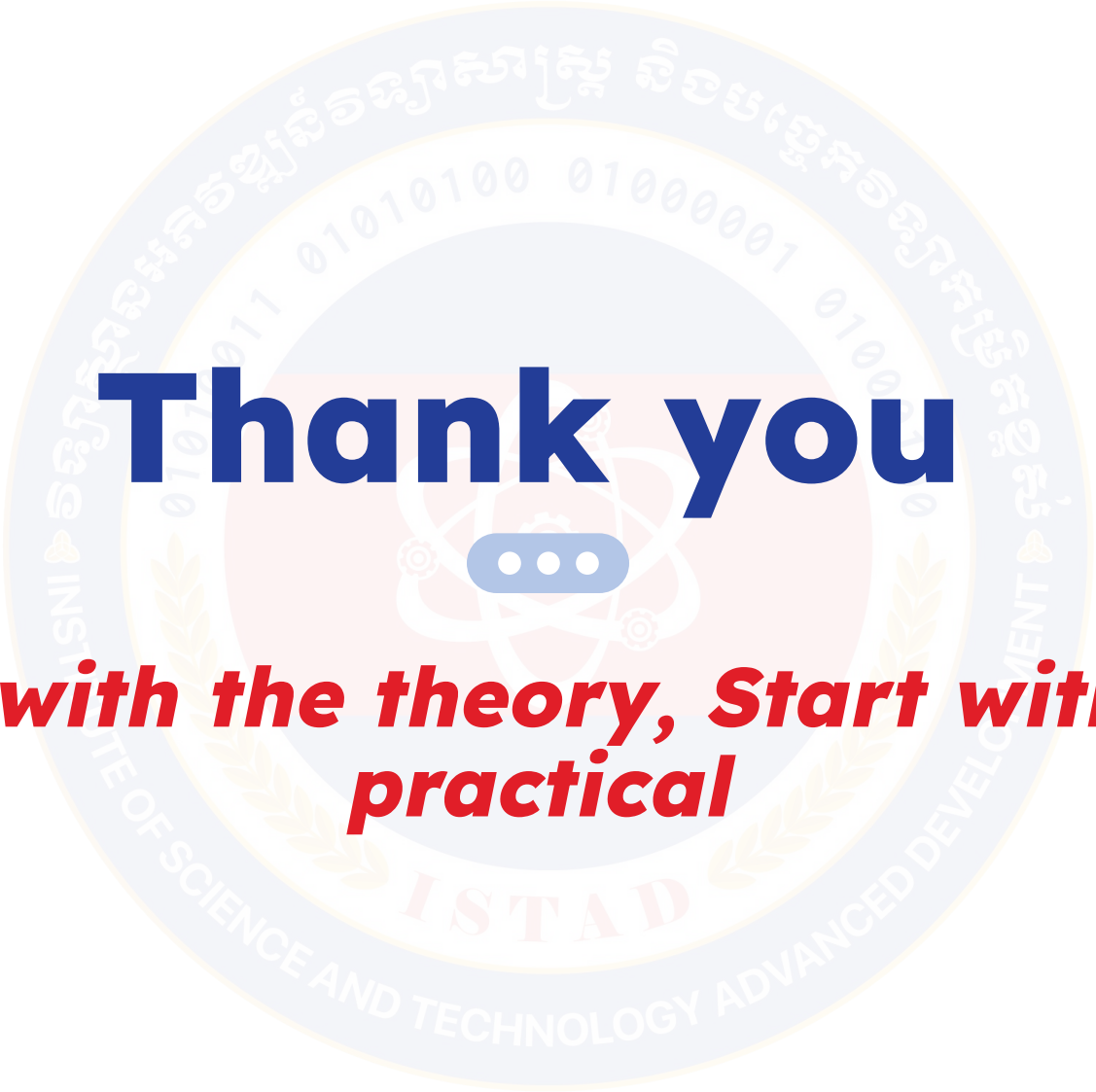
Group of Operating System

An Operating systems can be grouped into several categories based on various factors, including architecture, purpose, and user interface. Some common groups of operating systems include:

- **Desktop operating systems:** Designed for use on personal computers and include Microsoft Windows, Apple macOS, and Linux.
- **Server operating systems:** Designed to run on servers and provide services over a network, such as Microsoft Windows Server, Linux, and Unix.
- **Mobile operating systems:** Designed for use on mobile devices, such as smartphones and tablets, and include iOS, Android, and Windows Phone.

Group of Operating System

- **Embedded operating systems:** Designed for use in embedded devices, such as smart home devices, automotive systems, and medical equipment. Examples include Embedded Linux and RTOS (Real-Time Operating System).
- **Real-time operating systems:** Designed for use in systems that require a quick response time, such as avionics, industrial control systems, and robotics.
- **Mainframe operating systems:** Designed for use on large, centralized data centers and are used to handle large amounts of data and support many concurrent users. Examples include IBM z/OS and UNIX System Services.



Thank you

Begin with the theory, Start with the practical